



FIRST SEMESTER 2026-27

Course Handout Classical and Quantum Waves

Date: 1.08.2026

In addition to part I (the General Handout for all courses, appended to the Time table), this section provides further specific details regarding the course.

Course Number: PHY U101

Course Title: Classical and Quantum Waves

Instructor-in-Charge: Kaushar Vaidya

Instructors: Anshuman Dalvi, Raj Kumar Gupta, Tapomoy Guha Sarkar, Kaushar Vaidya

Tutorial Instructors: Debasish Bandyopadhyay, Arpan Das, Amol Holkundkar, Tanmoy Mondal, Sajal Mukherjee, D. D. Pant, Niladri Sarkar, Kaushar Vaidya

1. Course Description:

This course takes us from the study of oscillations to waves, and finally to the foundations of quantum mechanics, tracing a connected thread through classical and modern physics.

We begin with an introduction to oscillations and the model of simple harmonic oscillations, damped and forced oscillations, leading to the phenomenon of resonance. We shall study coupled oscillators: normal modes and frequencies, and the continuum limit of a large number of coupled oscillators. We will study transverse vibrations of a stretched string, the one-dimensional wave equation, and traveling wave solutions to the wave equation. We shall study energy transport in waves and pulses on a string, and their interaction with a boundary: reflection and refraction. Wave phenomena, such as the interference of light waves from two coherent sources, will lead us to wave-particle duality, the concept of the photon, the de Broglie hypothesis, and matter waves. We shall study the Heisenberg uncertainty principle, the Schrödinger equation, and the wavefunction. We shall study the quantum problem of a particle in a one-dimensional box, energy quantization, barrier tunneling, the transmission coefficient, and introduce scanning tunneling microscopy as an application of quantum tunneling.

2. Scope and Objective of the Course:

- **Understand oscillatory motion:** Analyze simple harmonic motion, superposition of oscillations, and the effects of damping and external driving forces, including resonance and quality factor
- **Master the physics of coupled systems and waves:** Explain how coupled oscillators lead to normal modes, and how taking the continuum limit gives rise to wave phenomena, including the wave equation, traveling waves, and energy transport
- **Analyze wave behavior and propagation:** Distinguish between group and phase velocity, and describe how waves interact with boundaries and each other through reflection, refraction, and interference
- **Grasp the foundations of quantum mechanics:** Explain wave-particle duality, the de Broglie hypothesis, and the Heisenberg uncertainty principle, and apply the Schrödinger equation to determine wavefunctions and



energy quantization in simple systems

- **Apply quantum tunneling to real-world technology:** Calculate transmission coefficients for barrier tunneling and connect this phenomenon to practical applications such as scanning tunneling microscopy

3. Text Books:

T1 Vibrations and Waves, A.P. French, CBS, 2003

T2 Concepts of Modern Physics by Arthur Beiser, Shobit Mahajan, and S. Rai Choudhury, Tata McGraw-Hill Companies, 2006

4. Reference Books:

1. R1: Physics Vol I & II, Halliday/Resnick/Krane 5th Edition, John Wiley, 2003
2. R2: Quantum Physics of Atoms, Molecules, Solids, Nuclei, and Particles, Eisberg & Resnick, John Wiley & Sons (2nd edition, 1985)

5. Course Learning Outcomes:

1. Develop the ability to analyze and understand oscillatory mechanical systems that are coupled
2. Ability to solve the wave equation for certain boundary conditions
3. To understand Interference and diffraction phenomena
4. To describe wave-particle duality and the photon concept, derive the de Broglie wavelength for matter waves, and articulate the physical meaning of the Heisenberg uncertainty principle
5. Solve the Schrödinger equation for a particle in a one-dimensional box, interpret the resulting wavefunctions, and calculate the quantized energy levels of the system
6. Calculate the transmission coefficient for a particle tunneling through a potential barrier, and explain how this phenomenon underlies the operation of a scanning tunneling microscope

6. Course Plan:

Lecture Numbers	Module	Learning Objectives	Topics to be covered	Reference
L1	Simple Harmonic Motion	• Familiarize with the SHM	Introduction of Oscillations, Potential Energy Minima, and Departure from stability	Chap 3 T1
L2 – L3	Damped Oscillations	• Analyze the SHM under viscous damping • Define Quality factor • Express Amplitude and Energy as a function of time and number of cycles for light damping • Write the solution for critical and heavy damping	Effect of Viscous Damping, Decay of Free Vibrations, Quality Parameter, Light Damping, Critical Damping, Heavy Damping	Chap 3 T1



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L4 – L6	Forced Oscillators	- Analyze the steady state and transient solutions for damped oscillators; Understand the steady state and transient solutions as a special case - Understand the amplitude resonance - Determine the resonant frequency for amplitude resonance - Read power-absorption graphs	Forced Oscillations, Amplitude Resonance, Transient Phenomena, Power Resonance Derivation (Tutorial)	Chap 4 T1
L7 – L9	Coupled Oscillators	- Write the coupled equations of motion - Determine normal mode frequencies using the special method and analytical method - Understand resonance in two forced coupled oscillators	Two Coupled Oscillators, Determination of Normal Frequencies (Special Method, Analytical Method), Forced Vibrations for Coupled Oscillators	Chap 5 T1
L10 – L12	N-Coupled Oscillators, Continuum Limit	- Derive N normal mode frequencies - Write particle displacements as a function of position and time - Determine the permitted frequencies of stationary vibrations - Describe the motion of the string - Analyze the effect of driving force on vibrations in a stretched string	N Coupled Oscillators, Properties of the Normal Modes of N Coupled Oscillators, Large N case, Continuous Systems, Free Vibrations of Stretched Strings, Forced Harmonic Vibrations of a Stretched String	Chap 5, Chap 6 T1
L13 – L16	Traveling Waves, Energy	Describe a progressing wave and write it mathematically	Traveling wave solutions; Progressing Waves in One Direction; Superposition; Wave	Chap 7 T1



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		- Analyze how two traveling waves combine to form an interference pattern and wave pulses - Understand dispersion	speed; Energy Density and Energy Transport; Power carried by a wave; Pulses on a string	
L17 – L18	Interference: The Double Slit Experiment	- Review the double-slit experiment - Understand the phenomenon of interference	Young's double-slit experiment; Interference fringes; Wave-Particle duality through Davisson-Germer & Tonomura single-electron double-slit experiments	Chap 8 T1, Chap 2 T2
L19 – L21	Wave-particle duality	- Analyze the photoelectric effect - Analyze the Compton scattering - Understand the de Broglie hypothesis - Derive and understand the Heisenberg uncertainty principle - Distinguish between the phase velocity and the group velocity of matter waves	Concept of Photon; Photoelectric effect; Compton scattering; Double-slit with single particles (electrons, neutrons); de Broglie hypothesis; Phase and Group velocity of matter waves; Heisenberg uncertainty principle from wave-packet	Chap 2, Chap 3 T2
L22 – L24	Schrödinger equation; Particle in a box	- Define wavefunction - Formulate the time-independent Schrödinger equation - Explain the zero-point energy and energy quantization - Interpret the correspondence principle - Describe the solution of the finite square well problem	Wavefunction and Born probability interpretation; Normalization; Time-independent Schrödinger equation; Infinite square well: Boundary conditions, Zero-point energy; Energy quantization; Expectation values; Correspondence principle; Finite square well (qualitative); Application: quantum dots as tunable particle-in-a-box	Chap 5 T2
L25 – L28	Barrier Tunneling; STM	Formulate the Schrödinger equation for a rectangular potential barrier	Rectangular potential barrier; Wavefunction matching at boundaries; Transmission coefficient; Exponential	Chap 5 T2



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		• Apply continuity conditions on the wavefunction and its derivatives at boundaries • Interpret quantum tunneling through the potential barrier • Explain the working principle of the STM	sensitivity to barrier width and height; Scanning tunneling microscope: Tunnel current, Constant-current mode, Atomic resolution; Experiment: IBM quantum corral	
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7. Evaluation Scheme:

Component	Duration	Marks/Weightage	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	30%		Open
Continuous evaluation (Tutorial Test)	25 Min.	30%	Throughout	Closed
Comprehensive Examination	180 Min.	40%		Closed

8. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

9. NC Criteria: 30% of the average of the top three performers or 40% of the median, whichever is lower.

10. Chamber Consultation Hour: TBA

11. Notices: On Google Classroom and Nalanda

12. Make-up Policy: Very Strict. Only in the most genuine cases, as judged by the IC.

13. Note (if any):

Instructor-in-charge
PHY U101